Hojoon Leo Kim

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Research Interests

Machine Learning Systems, Hardware–Software Co-Design, Energy-Efficient AI

My research focuses on agent–hardware co-design: jointly optimizing agentic algorithms for planning and memory with accelerator architectures and execution mechanisms such as plan reuse. I extend this full-stack approach to embodied AI and robotics, where on-device vision–language–action (VLA) models require real-time, energy-efficient inference. My long-term goal is to develop reusable system abstractions and open benchmarks that bridge ML algorithms and accelerator architecture, enabling scalable deployment of agentic and embodied AI.

Education

- **Seoul National University** — *Seoul, South Korea*
  M.S. in Computer Science and Engineering (Advisor: Professor Jae W. Lee) — Sep 2026 – Aug 2028 (Expected)
  B.S. in Electrical and Computer Engineering, Summa Cum Laude — Mar 2020 – Feb 2026
  - B.S. GPA: 3.99/4.30 (Major GPA: 4.10/4.30)
- **Stanford University** — *Jun 2025 – Aug 2025*
  Visiting Student Researcher — Stanford, CA, USA

Publications

C=Conference, S=In Submission

* Equal contribution

[C.3] Hojoon Kim, Yuheng Wu, and Thierry Tambe. **AgenticCache: Cache-Driven Asynchronous Planning for Embodied AI Agents.** *MLSys'26*. (Acceptance Rate: 26.4%) [Paper PDF]

[C.2] Seung Yul Lee, Hojoon Kim, Yutack Park, Dawoon Jeong, Seungwu Han, Yeonhong Park, and Jae W. Lee. **FlashTP: Fused, Sparsity-Aware Tensor Product for Machine Learning Interatomic Potentials.** *ICML'25 Spotlight* (Spotlight Rate: 2.6%) [Paper PDF]

[C.1] Yeonhong Park*, Jake Hyun*, Hojoon Kim, and Jae W. Lee. **DecDEC: A Systems Approach to Advancing Low-Bit LLM Quantization.** *OSDI'25*. (Acceptance Rate: 15.9%) [Paper PDF]

[S.4] Hojoon Kim, Chanyeong Lim, Jae W. Lee, and Kiwan Maeng. **Carnevale: Hardware-Software Co-Design for Efficient Masked Machine Learning at Scale.** *In submission to ASPLOS'27*

[S.3] Jaehoon Yang, Yongbeom Kim, Hojoon Kim, Seung Yul Lee, and Jae W. Lee. **FLIT: A Fine-Grained Memory Management Layer for Colocating LLM Inference and Tuning.** *In submission to EuroSys'27*

[S.2] Jongheon Jeong*, Hojoon Kim*, Rokhee Lee, Yeonhong Park, Young H. Oh, and Jae W. Lee. **Helios: Heterogeneous lightweight VLA model serving system.** *In submission to ASPLOS'27*

[S.1] Seong Hoon Seo, Donghyun Lee, Geonha Lee, Hojoon Kim, Yeonhong Park, and Jae W. Lee. **QUESO: Storage-Assisted Quantization Error Compensation for On-Device LLM Inference.** *In submission to DATE'27*

Research Experience

- **SNU Architecture and Code Optimization Lab [ARC Lab]** — *Seoul, South Korea*
  Graduate Researcher (P.I.: Professor Jae W. Lee) — Feb 2026 – Aug 2028 (Expected)
  Undergraduate Researcher — Sep 2024 – Feb 2026
  - Led the compiler–architecture co-design for exact INT64 arithmetic, developing optimized GPU kernels and proposing GPU microarchitectural extensions for scalable masked machine learning. (ASPLOS'27 submission)
  - Led the design and implementation of HELIOS, a lightweight GPU kernel orchestrator for centralized multi-robot VLA serving. (ASPLOS'27 submission)
- **Stanford Tambe Lab [Tambe Lab]** — *Jun 2025 – Mar 2026*
  Undergraduate Researcher (P.I.: Professor Thierry Tambe) — Stanford, CA
  - Led research on cache-driven asynchronous planning for embodied AI agents by exploiting plan-level locality. (MLSys'26)
- **SNU Computer Architecture and Systems Lab [COMPARCH Lab]** — *Dec 2023 – Sep 2024*
  Research Intern (P.I.: Professor Jaewoong Sim) — Seoul, South Korea
  - Led undergraduate thesis based on Tender (ISCA'24) quantization architecture; designed hardware module to shift offline computation to runtime, improving model accuracy.

Academic Projects

- **A File System with Full-Path Indexing: Operating Systems (1st Place)** *Dec 2024*
- **Pipelined Central Processing Unit with Perceptron Branch Predictor: Computer Organization** *Jun 2024*
- **L1, L2, and Memory Three-Level Memory Simulator with MSHR: Computer Organization** *Jun 2024*
- **A CNN Accelerator on FPGA (Artix-7): Digital Systems Design and Lab (1st Place)** *Dec 2023*

Honors and Awards

- **YoulChon AI Young Researcher Scholarship** *Sep 2026*
  Yulchon Foundation & Seoul National University AI Institute
- **Semiconductor Specialized Scholarship** *Jan 2024 – Feb 2026*
  Semiconductor Specialization Project Group, Seoul National University
- **Grand Prize – 2025 AI Chip Contest - NPU Optimization Track** *Nov 2025*
  Ministry of Science and ICT (MSIT), Republic of Korea
- **Global Leadership Program (airfare)** *Jun 2025 – Aug 2025*
  College of Engineering
- **Merit-Based Tuition Scholarship** *Aug 2023 – Jun 2025*
  Department of Electrical and Computer Engineering
- **1st Prize – Advanced CUDA Accelerator Programming School** *Feb 2025*
  SNU Thunder Research Group
- **4th Prize – Military AI Contest** *Dec 2022*
  Ministry of National Defense & Ministry of Science and ICT

Academic Service

- **Program Committee Member, MLSys 2027** *2027*
  Conference on Machine Learning and Systems

Teaching Experience

- **Peer Tutor, Computer Organization; Programming Methodology** *Sep 2024 – Jun 2025*
  Department of Electrical and Computer Engineering

Military Service

- **SIGINT Specialist, Republic of Korea Army** *Jan 2022 – Jul 2023*
  - Two-time recipient of the Best SIGINT Analyst Award; recognized as a First-Class Soldier (fitness, mental readiness, and specialty proficiency).

Extracurricular Activities

- **Member, SNU Business School Tennis Club** *Mar 2024 – Dec 2024*
- **Peer Mentor, SNU Center for Campus Life & Culture** *Mar 2024 – Jun 2024*
  - Received the Best Mentor Award and Best Mentoring Team Award.

Technical Skills

- **Programming Languages:** C/C++, Python, Verilog/SystemVerilog
- **ML Systems and GPU Programming:** CUDA, PyTorch, NVIDIA Nsight Systems/Compute
- **EDA and FPGA Tools:** Synopsys Design Compiler (logic synthesis, area estimation), Synopsys PrimeTime (timing/power analysis), Xilinx Vivado
- **Architecture Simulators:** Accel-Sim, GPGPU-Sim

Language Proficiency

Korean (Native), English (TOEFL 106 [R:30/L:28/S:22/W:26], OPIc AL)